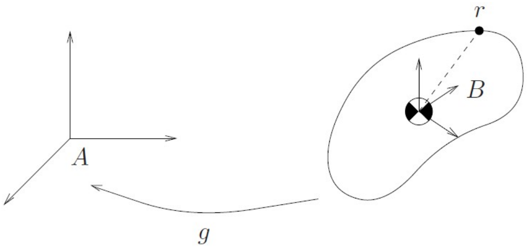
Chapter 4 §2.2 Inertial properties of rigid bodies

Consider a rigid body as shown in MLS figure 4.2, below.

MLS p. 161:

$$v_r^s = \dot{p} + \dot{R}r$$



We now denote the centre of mass of the body as G.

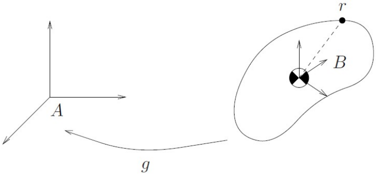
The body frame B origin is located at G.

Angular momentum about G

Consider a rigid body as shown in MLS figure 4.2, below.

MLS p. 161:

$$v_r^s = \dot{p} + \dot{R} r$$



Velocity of r relative to G:

$$v_{r/G}^s = \dot{R} r$$

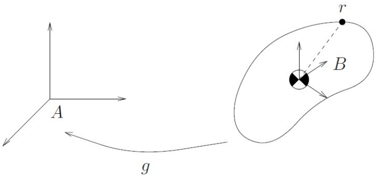
$$v_{r/G}^b = R^T v_{r/G}^s = R^T \dot{R} r = \hat{\omega} r$$

where $\omega \in \mathbb{R}^3$ is the angular velocity of the body in the body frame (middle p. 162).

Angular momentum about G

The angular momentum of the body point r about the body centre of mass, relative to the body centre of mass, expressed in the body frame, is

$$
\begin{aligned}
dH^b_{G/G} &= \rho(r)\hat{r}v^b_{r/G}\,dV \\
&= \rho(r)\hat{r}(\hat{\omega}r)\,dV \\
&= -\rho(r)\hat{r}^2\omega\,dV
\end{aligned}
$$



Therefore,

$$H^b_{G/G} = -\int_V \rho(r)\hat{r}^2\,dV\omega = \mathcal{I}\omega$$

where $\mathcal{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} = -\int_V \rho(r)\hat{r}^2\,dV$ (see p. 162) and moment_of_inertia_tensor.pdf

Angular momentum about G

$$H^b_{G/G} = \mathcal{I}\omega$$

$$H^s_{G/G} = RH^b_{G/G} = R\mathcal{I}\omega = R\mathcal{I}R^T R\omega = (R\mathcal{I}R^T)(R\omega)$$

$$H^s_{G/G} = \mathcal{I}'\omega^s$$

where

$$\mathcal{I}' = R\mathcal{I}R^T \qquad \text{and} \qquad \omega^s = R\omega$$

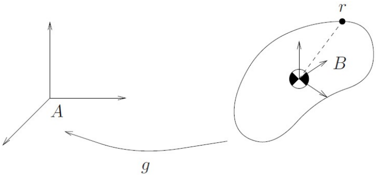
This is used in the middle of p. 166 in the derivation of the Euler's equation.

Kinetic energy

Consider a rigid body as shown in MLS figure 4.2, below.

MLS p. 161:

$$v_r^s = \dot{p} + \dot{R}r$$



$$T = \frac{1}{2}\int_V \rho(r)||v_r^s||^2 \, dV$$

$$\begin{aligned} T &= \frac{1}{2}\int_V \rho(r)(\dot{p} + \dot{R}r)^T(\dot{p} + \dot{R}r) \, dV \\ &= \frac{1}{2}\int_V \rho(r)\left(||\dot{p}||^2 + 2\dot{p}^T\dot{R}r + ||\dot{R}r||^2\right) \, dV \end{aligned}$$

Kinetic energy

$$T = \frac{1}{2}\int_V \rho(r)\left(||\dot{p}||^2 + 2\dot{p}^T\dot{R}r + ||\dot{R}r||^2\right)\ dV$$

- $\frac{1}{2}\int_V \rho(r)||\dot{p}||^2\ dV = \frac{1}{2}\int_V \rho(r)\ dV||\dot{p}||^2 = \frac{1}{2}m||\dot{p}||^2$, where m is the mass of the body (p. 161); translational kinetic energy
- $\int_V \rho(r)\dot{p}^T\dot{R}r\ dV = \dot{p}^T\dot{R}\int_V \rho(r)r\ dV = 0$, as B lies on G.
- 3rd term needs further analysis.

Kinetic energy

$$T = \frac{1}{2} \int_V \rho(r) \left(||\dot{p}||^2 + 2\dot{p}^T \dot{R} r + ||\dot{R} r||^2 \right) \, dV$$

$$\begin{aligned}
\frac{1}{2} \int_V \rho(r) ||\dot{R} r||^2 \, dV &= \frac{1}{2} \int_V \rho(r) (\dot{R} r)^T (\dot{R} r) \, dV \\
&= \frac{1}{2} \int_V \rho(r) (R R^T \dot{R} r)^T (R R^T \dot{R} r) \, dV \\
&= \frac{1}{2} \int_V \rho(r) (R \hat{\omega} r)^T (R \hat{\omega} r) \, dV \\
&= \frac{1}{2} \int_V \rho(r) (R \hat{r} \omega)^T (R \hat{r} \omega) \, dV
\end{aligned}$$

Kinetic energy

$$
\begin{aligned}
\frac{1}{2}\int_V \rho(r)||\dot{R}r||^2\, dV &= \frac{1}{2}\int_V \rho(r)(R\hat{r}\omega)^T(R\hat{r}\omega)\, dV \\
&= \frac{1}{2}\int_V \rho(r)\omega^T\hat{r}^T R^T R\hat{r}\omega\, dV \\
&= \frac{1}{2}\omega^T\int_V \rho(r)\hat{r}^T\hat{r}\, dV\omega \\
&= -\frac{1}{2}\omega^T\int_V \rho(r)\hat{r}^2\, dV\omega \\
&= \frac{1}{2}\omega^T\mathcal{I}\omega
\end{aligned}
$$

Kinetic energy

$$\text{For } \hat{V}^b = g^{-1}\dot{g} \in se(3): \quad V^b = \begin{bmatrix} R^T \dot{p} \\ \omega \end{bmatrix}$$

(see MLS eq. (2.55) p. 55)

Consider now

$$\begin{aligned} V^{bT} \begin{bmatrix} mI & 0 \\ 0 & \mathcal{I} \end{bmatrix} V^b &= m(R^T\dot{p})^T(R^T\dot{p}) + \omega^T \mathcal{I} \omega \\ &= m||R^T\dot{p}||^2 + \omega^T \mathcal{I}\omega \\ &= m||\dot{p}||^2 + \omega^T \mathcal{I}\omega \end{aligned}$$

Kinetic energy

$$T = \frac{1}{2}\int_V \rho(r)\left(||\dot{p}||^2 + 2\dot{p}^T\dot{R}r + ||\dot{R}r||^2\right)\; dV$$

$$\begin{aligned} T &= \frac{1}{2}m||\dot{p}||^2 + \frac{1}{2}\omega^T\mathcal{I}\omega \\ &= \frac{1}{2}V^{bT}\begin{bmatrix} mI & 0 \\ 0 & \mathcal{I} \end{bmatrix}V^b = \frac{1}{2}V^{bT}\mathcal{M}V^b \qquad (4.9) \end{aligned}$$

$\mathcal{M}$: generalized inertia matrix expressed in body frame, symmetric and positive definite.

§2.4 Newton-Euler equations for rigid bodies

Linear momentum of body: $m\dot{p}$.

Newton II for linear motion:

$$f = m\frac{d}{dt}(\dot{p})$$

Mass constant:

$$f = m\ddot{p} \tag{4.12}$$

Independent of angular motion as G is at B.

§2.4 Newton-Euler equations for rigid bodies

Newton II for angular motion, for $\tau \in \mathbb{R}^3$ in inertial frame:

$$\tau = \frac{d}{dt} H^s_{G/G} = \frac{d}{dt}(\mathcal{I}'\omega^s) = \frac{d}{dt}(R\mathcal{I}R^T\omega^s)$$

Therefore

$$\begin{aligned}
\tau &= R\mathcal{I}R^T\dot{\omega}^s + \dot{R}\mathcal{I}R^T\omega^s + R\mathcal{I}\dot{R}^T\omega^s \\
&= \mathcal{I}'\dot{\omega}^s + \dot{R}R^T R\mathcal{I}R^T\omega^s + R\mathcal{I}R^T R\dot{R}^T\omega^s \\
&= \mathcal{I}'\dot{\omega}^s + \dot{R}R^T\mathcal{I}'\omega^s + \mathcal{I}'R\dot{R}^T\omega^s \\
&= \mathcal{I}'\dot{\omega}^s + \hat{\omega}^s\mathcal{I}'\omega^s - \mathcal{I}'\hat{\omega}^s\omega^s \\
&= \mathcal{I}'\dot{\omega}^s + \hat{\omega}^s\mathcal{I}'\omega^s \qquad (4.13)
\end{aligned}$$

§2.4 Newton-Euler equations for rigid bodies

$$f = m\ddot{p} \tag{4.12}$$

$$\tau = \mathcal{I}'\dot{\omega}^s + \hat{\omega}^s\mathcal{I}'\omega^s \tag{4.13}$$

Euler's equation (expressed in inertial frame)

$(f, \tau) \in \mathbb{R}^6$ is not a wrench applied to the rigid body in the sense of chapter 2, as it is not applied at the inertial frame origin A.

$(\dot{p}, \omega^s) \in \mathbb{R}^6$ is not the spatial velocity ($\dot{p}$ is wrong)

§2.4 Newton-Euler equations for rigid bodies

To express the dynamics in terms of twists and wrenches, we need to use the body velocity $v^b = R^T \dot{p}$ and body force $f^b = R^T f$.

$$f = m\ddot{p} = m\frac{d}{dt}(\dot{p}) = m\frac{d}{dt}(Rv^b) = Rm\dot{v}^b + \dot{R}mv^b$$

$$R^T f = R^T Rm\dot{v}^b + R^T \dot{R}mv^b$$

$$f^b = m\dot{v}^b + \hat{\omega}mv^b \tag{4.14}$$

$$\tau = \mathcal{I}'\dot{\omega}^s + \hat{\omega}^s\mathcal{I}'\omega^s \tag{4.13}$$

$$R^T\tau = R^T\mathcal{I}'RR^T\dot{\omega}^s + R^T\hat{\omega}^s RR^T\mathcal{I}'RR^T\omega^s$$

$$\tau^b = \mathcal{I}\dot{\omega} + \hat{\omega}\mathcal{I}\omega \tag{4.15}$$

§2.4 Newton-Euler equations for rigid bodies

$$f^b = m\dot{v}^b + \hat{\omega} m v^b \tag{4.14}$$

$$\tau^b = \mathcal{I}\dot{\omega} + \hat{\omega}\mathcal{I}\omega \tag{4.15}$$

$$F^b = \begin{bmatrix} mI & 0 \\ 0 & \mathcal{I} \end{bmatrix} \begin{bmatrix} \dot{v}^b \\ \dot{\omega} \end{bmatrix} + \begin{bmatrix} m\hat{\omega} v^b \\ \hat{\omega}\mathcal{I}\omega \end{bmatrix} \tag{4.16}$$

Newton-Euler equations in body frame components.

F^b is an external wrench applied at G, expressed in body frame components.